Qualcomm Technologies, Inc.

QTI Test Tools – SW6100 Wearable Development Platform Overview

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

80-74841-118 Rev. AC

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets

NO PUBLIC DISCLOSURE PERMITTED: Please report postings of this document on public servers or websites to: DocCtrlAgent@qualcomm.com.

© Qualcomm Technologies, Inc. and/or its subsidiaries. All rights reserved.

Revision History

Revision	Date	Description
AA	August 2025	Initial release
AB	October 2025	Updated slides 12, 13, 14, and 15
AC	November 2025	Updated slides 10 and 23

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

Contents

1 SW6100 Wearable Development Platform (WDP) Overview 4

2 WDP SW6100 Test Device 9

3 WDP SW6100 Software Overview 16

4 Regulatory Notice 18

5 Purchase WDP 20

6 References 22

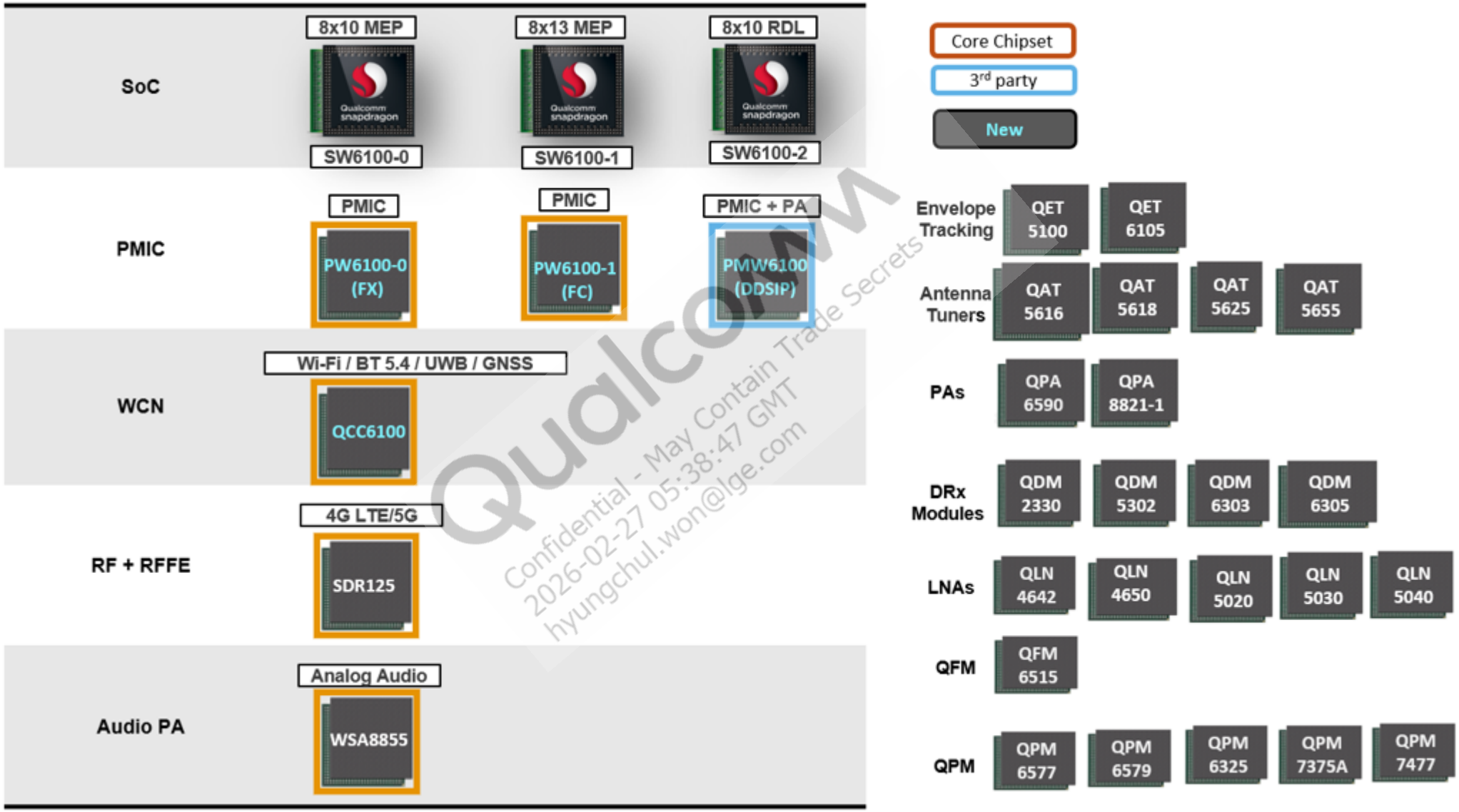
Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

Section 1

SW6100 Wearable Development Platform (WDP) Overview

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

SW6100 Chipset Summary with SDR125



Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

SW6100 Device Description

The SW6100-0/SW6100P-0 is a 3 nm process chip, with a next-generation Qualcomm Snapdragon® processor and LTE modem supporting 4G/5G sub-6 technologies crafted exclusively for wearable designs.

Key processor and memory characteristics include:

- Arm Cortex-A78 applications processor supporting up to 2.1 GHz, with 256 kB L2 cache
- Quad Core Arm Cortex-A55 applications processor supporting up to 1.95 GHz, with 64 kB L2 dedicated cache per core
- 3 MB of shared L3 cache between all the cores
- Qualcomm® Adreno™ 621 graphics processing unit (GPU)
- Dedicated 2.5D LPI GPU for always-on display (AoD)
- Integrated wearable microcontroller (WM) system with Dual Arm Cortex
- M55 core and display co-processor (DCP) to handle low-power display, audio, and sensor use cases
- Dedicated DSP for Snapdragon sensor core and low-power audio subsystem
- Dual eNPU for high-performance and always-on neural network (NN) use cases
- Dedicated NSP for AI applications
- Package-on-package (PoP) high-speed eMCP memory:
  - 1 × 16 3200/4266 MHz LPDDR5/5X SDRAM up to 4 GB
  - eMMC 5.1 up to 64 GB
- Always-on subsystem with RPM for power management

NOTE: The features listed on this slide denote SW6100 mobile platform capabilities; the WDP support for these features depends on the hardware capability and the software support. The features listed on this slide are preliminary and subject to change.

Qualcomm Confidential – May Contain Trade Secrets 2026-02-27 05:38:47 GMT hyungchul.won@lge.com

SW6100 Device Description (cont.)

Key features include:

- Qualcomm® Universal Bandwidth Compression (UBWC) with GPU
- 720 × 720 60 Hz, 4-layers, 1 × 4-lane DSI (D-PHY v1.2)
- 13 + 13 MP, 25 MP, 1080p60, 1 × 4-lane (D-PHY 1.2/C-PHY 1.2)
- 1 × eUSB2 support, with repeater integrated inside the PMIC
- PCIe Gen3 1-lane
- 1.2 V/1.8 V capable I/Os
- Multiple QUP engines to support I^2C/UART/SPI/I^3C
- Support for PSRAM with Hexadecimal/Octal SPI interface
- Supports 3GPP Rel.17 Red. Cap with multimode 4G and 5G sub-6
- SA transceiver SDR125
- QCC6100 chip attach to SoC offering:
  - WLAN IEEE 802.11a/b/g/n/ac/ax with Tri-band (2.4/5/6GHz)
  - Bluetooth® v5.4 with Bluetooth low energy (BLE) audio
  - L1 and L5 GNSS support for navigation applications
  - UWB: 1 Tx + 1 Rx configuration

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

NOTE: The features listed on this slide denote SW6100 mobile platform capabilities; the WDP support for these features depends on the hardware capability and the software support. The features listed on this slide are preliminary and subject to change.

Key Modem Features in SW6100 Mobile Platform

Technology	Features	SW6100 support
3GPP compliance		3GPP Rel. 17 Redcap 5G NR Sub-6
WAN	Radio Access Tech	LTE Cat4 + 5G Release 17 RedCap ■ Peak data rates (NR): 223 Mbps DL/123 Mbps UL ■ Peak data rates (LTE): 200 Mbps DL/105 Mbps UL
RF Bands	LTE, NR bands	NR bands NR bands – 600 MHz to 7 GHz, LTE bands 600 MHz to 3.7 GHz
NTN	Non-terrestrial network	NB-NTN
LTE PHY	DL Modulation	QPSK, 16 QAM, 64 QAM, and 256 QAM
	UL Modulation	QPSK, 16 QAM, 64 QAM, and 256 QAM
	Bandwidth DL/UL	20 MHz, 1CC
LTE Features	Cat 1bis	LTE Release 16, Cat 1bis (Cat1 with single Rx)
	Cat 4	LTE Cat4 – 2Rx, 1Rx
	Voice	VoLTE, IMS emergency call, and robust header compression (RoHC) (AMR/EVS)
5G Features	Release 17	5G NR SA mode only, iRAT to LTE
	Release 17 NR RedCap TDD/FDD	Coverage enhancement and power-saving features from Release 17 RedCap WI
	Duplex	TDD and FDD
	DL modulation	QPSK, 16 QAM, 64 QAM, and 256 QAM
	Bandwidth	20 MHz, 1CC
	Peak DL data rate	223 Mbps
	UL modulation	QPSK, 16 QAM, 64 QAM, and 256 QAM
	Peak UL data rate	123 Mbps
	Voice	VoNR, VoWiFi, and EPS fallback
Power	Power Features	WUS enhancements
		DCI-based BWP
		Release 17 power saving features (PEI, SSSG Switching)

Qualcomm
Confidential – May Contain Trade Secrets
2026-03-27 05:36:47 GMT
hyung.chunwoo@lge.com

Section 2

WDP SW6100 Test Device

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

WDP SW6100 – RF Variants and Band Support

WDP variant	WCDMA	LTE	5G Sub-6	Satellite (NTN)
SDR125-Global (MPSS.CY.2.0)	B1,B2,B4,B5,B6, B8,B19	B1,B2,B3,B4,B5,B7,B8,B11,B12,B13,B14, B17,B18,B19,B20,B21,B25,B26,B28,B30,B34,B38,B39, B40,B41,B42,B43,B48,B66,B68,B70,B71	n1,n2,n3,n5,n7,n8,n12,n13,n14,n18,n20, n25,n26,n28,n30,n38,n40,n41,n48,n66, n70,n71,n77,n78,n79,n91,n92,n93,n94	B253,B255,B256

Qualcomm
Confidential - May Contain Trade Secrets
2025-02-27 05:38:47 GMT
hyeonbul.won@lge.com

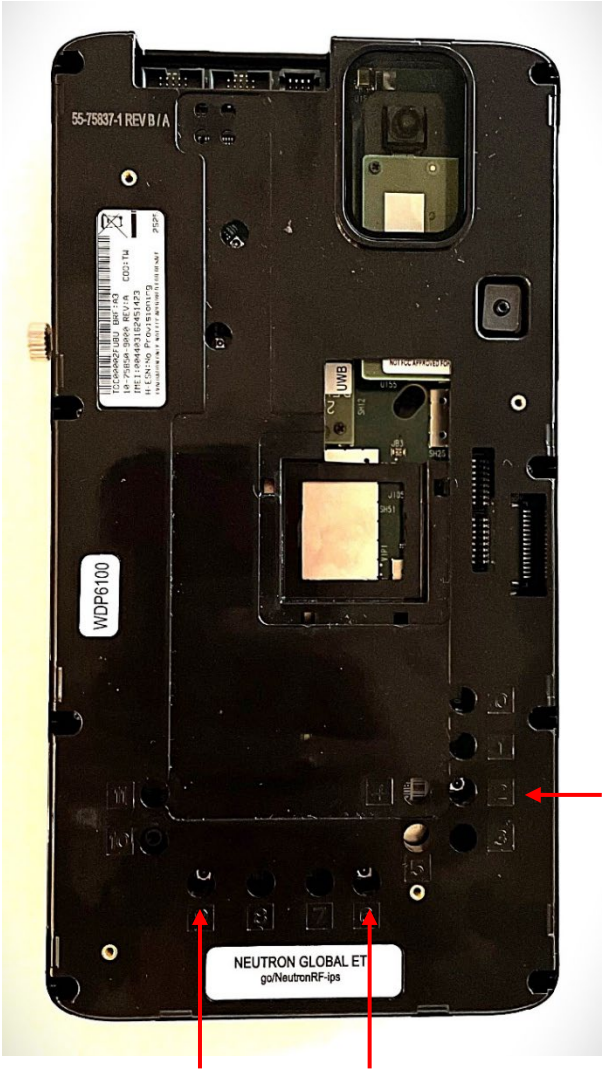
NOTES:

- This table represents the bands that the WDP SW6100 device supports. LTE Carrier Aggregation and 5G EN-DC are not supported in this device. The support for specific combinations is subject to change based on the 3GPP standardization, infrastructure support, and operator requirements.
- The listed bands on this slide denotes hardware capabilities; the band support depends on the software plan and is subject to change.
- Bands marked (*) may be disabled by default and require additional licensing.
- 450 MHz and GPS features may be disabled by default and require additional licensing.
- See *SW6100 + SDR125 RF Features Spreadsheet* (80-62400-650) for more information.

WDP6100 RF Port Map

Antenna	LB (MHz)	MHB (MHZ)	GPS(L2/L5) (MHz)	GPS(L1) (MHz)	Wi-Fi-2.5G (MHz)	5G-L (MHz)	5G-MH (MHz)	wi-Fi-5G (MHz)	UWB Channels (MHz)
Ant0	617-960	1427-2690				3300-4200	4400-5000		
Ant1			1166-1229	1565-1610					
Ant2	617-960	1427-2690			2400-2500	3300-4200	4400-5000		
Ant8					2400-2500			5150 – 7125	
Ant9			1166-1229	1565-1610					6500; 8000; 8500; 9000

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.woo@lge.com

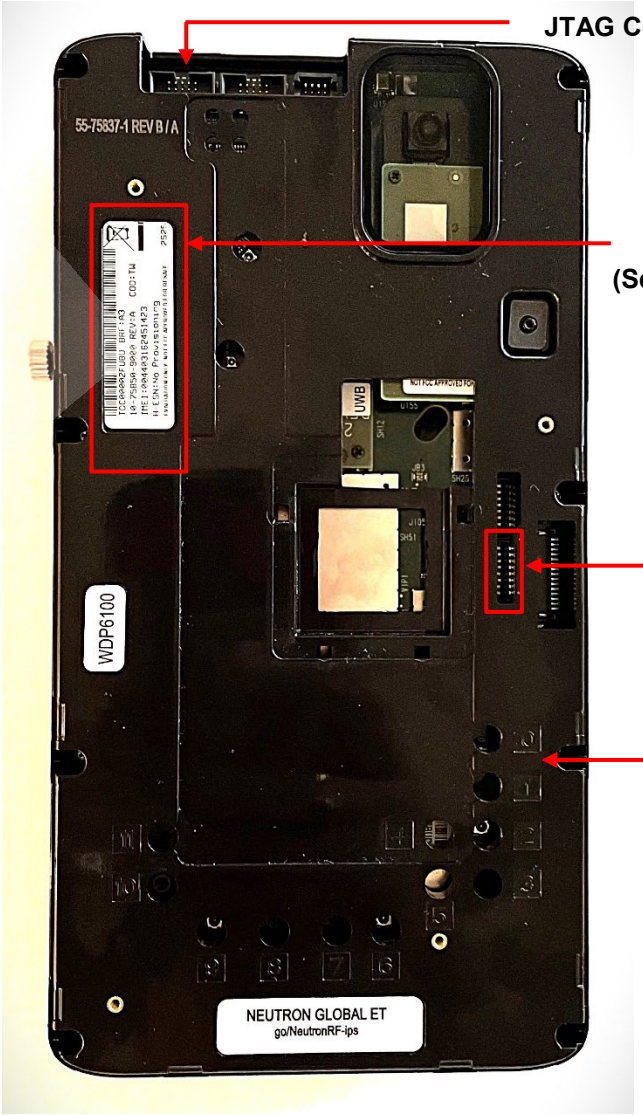


ANT 0

ANT 1 ANT 2

WDP SW6100 – Dimensions and Interfaces

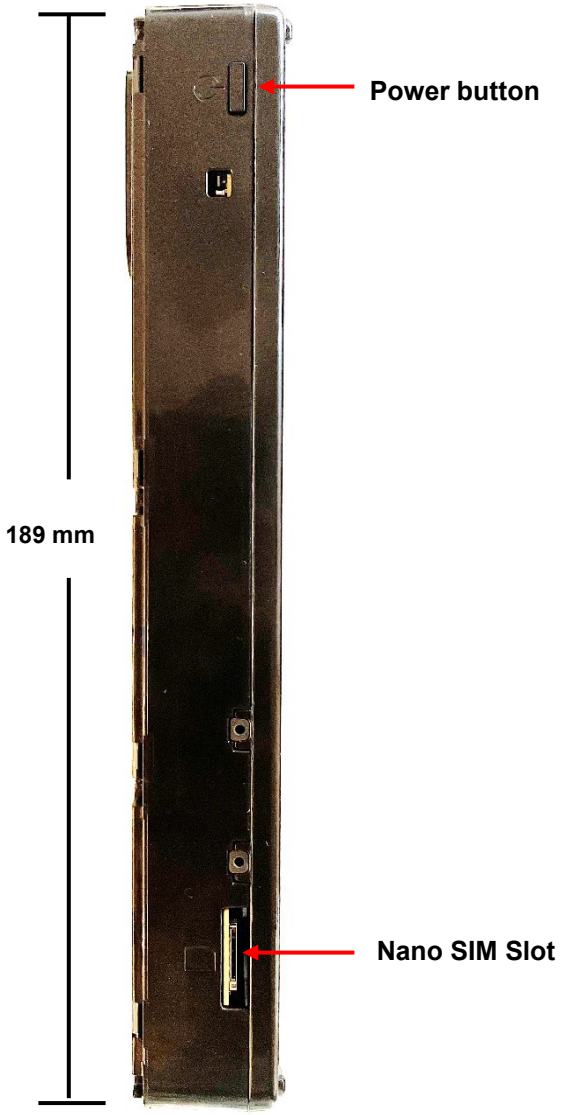
Front View



Back View

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

WDP SW6100 – Dimensions and Interfaces (cont.)



Side View

Top View

Bottom View

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

WDP SW6100 – Kit Contents

- WDP SW6100
  - SW6100 v1.0 – MCN 65-75850-9105
- Mini-JTAG Cable
- Rear housing
- USB Cable
- Power Adapter
- Battery
- Battery holder
- Wireless Charger Adapter

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

WDP SW6100 – MCNs and Alternate Device Type

- WDP SW6100 MCNs
  - SW6100 8×10 MEP with Physical SIM – MCN 65-75850-9105 (subject of this document)
  - SW6100 8×10 MEP – MCN 65-75850-xxxx
  - SW6100 8×13 MEP – MCN 65-85386-xxxx
  - SW6100 8×10 RDL – MCN 65-75850-9000
  - SW6100 8×10 RDL with PMW6100 – 65-85388-xxxx
- Alternate Device Type
  - Wearable Reference Design (WRD) SW6100
  - SKU1: 65-82482-xxxx SKU2: 65-82483-xxxx

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

NOTE: MCNs marked with an x are subject to change.

Section 3

WDP SW6100 Software Overview

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

WDP SW6100 Software Overview

- The software for this device has two release distributions available from the SW6100 Android test device product kit at CreatePoint:
  - Test device distribution: This distribution requires compilation of the Android open source.
    For example, SW6100-LAW-1-0 TEST DEVICE.
  - High-level operating system (HLOS) device OSS distribution: This distribution includes precompiled images ready to install to the WDP.
    For example, SW6100-LAW-1-0 HLOS DEVICE OSS. See *WDP SW6100 Software Installation and Configuration Guide* (80-74841-119), for more information.

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

NOTE: See the latest document revision at https://createpoint.qti.qualcomm.com.

Section 4

Regulatory Notice

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

Regulatory Notice

Regulatory Development device notice

This development device contains RF/digital hardware and software intended for engineering development, engineering evaluation, or demonstration purposes only and is intended for use in a controlled environment. This device is not being placed on the market, leased or sold for use in a residential environment or for use by the general public as an end-user device.

This development device is not intended to meet the requirements of a commercially available consumer device including those requirements specified in the European Union directives applicable for Radio devices being placed on the market, FCC equipment authorization rules or other regulations pertaining to consumer devices being placed on the market for use by the general public.

This development device may only be used in a controlled user environment where operators have obtained the necessary regulatory approvals for experimentation using a radio device and have appropriate technical training. The device may not be used by members of the general population or other individuals that have not been instructed on methods for conducting controlled experiments and taking necessary precautions for preventing harmful interference and minimizing RF exposure risks. The device should be installed and operated with a minimum distance of 8 inches (20 cm) between radiating antennas and bystanders. Additional RF exposure information can be found on the FCC website at http://www.fcc.gov/encyclopedia/radio-frequency-safety

Specific Regional Notices

European Union Regulatory Compliance. With respect to any Equipment or Hardware Development Tools (collectively, "**Hardware**") delivered to the European Union, upon the earlier of the end of the useful life of the Hardware, or when Buyer ceases to use any Hardware delivered under this Agreement, Buyer shall, at its cost, return the Hardware to QTI for proper disposal and in no event shall any Hardware be resold or placed on the market in any manner. Buyer will not operate the RF transmitter over the air (OTA) associated with any Test Tools in the European Union unless Buyer secures from the appropriate local governmental body the necessary authority to operate the transmitter.

U.S. Federal Communications Commission ("FCC") Regulatory Compliance. Buyer will not operate the RF transmitter over the air (OTA) associated with any Test Tool in the U.S. unless Buyer secures from the FCC the appropriate experimental authority or special temporary authorization pursuant to the U.S. Federal Communications Rules in the Code of Federal Regulations, Title 47, Part 5.

Additional FCC NOTICE: This kit is designed to allow:

Product developers to evaluate electronic components, circuitry, or software associated with the kit to determine whether to incorporate such items in a finished product and

Software developers to write software applications for use with the end product. This kit is not a finished product and when assembled may not be resold or otherwise marketed unless all required FCC equipment authorizations are first obtained. Operation is subject to the condition that this product not cause harmful interference to licensed radio stations and that this product accept harmful interference. Unless the assembled kit is designed to operate under part 15, part 18 or part 95 of this chapter, the operator of the kit must operate under the authority of an FCC license holder or must secure an experimental authorization under part 5 of this chapter.

If the kit is modified through the addition of external RF components (for example, power amplifiers or high gain antennas) then the owner must ensure operation with existing rules.

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:36:47 GMT
hyungchul.yoon@lge.com

Section 5

Purchase WDP

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

Purchase WDP

- The sales center is the web store where customers can purchase test tools like the WDPs and QXDM Professional™ Tool. See https://cp.qti.qualcomm.com for more details.

- Additional information on this product is available at https://createpoint.qti.qualcomm.com.

Section 6

References

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

Documents and Standards

Title	Number
Qualcomm Technologies, Inc.	
SW6100 Software User Guide	80-74841-50
WDP SW6100 Software Installation and Configuration Guide	80-74841-119
Snapdragon WRD6100 Wearable Reference Design Product Brief	87-82482-1
SW6100 SDR125 RF Features Spreadsheet	80-62400-650

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

Acronyms and Terms

Acronym or Term	Definition
ASC	Always sensing camera
CDMA	Code division multiple access
DLCA	Downlink carrier aggregation
DPU	Display processing unit
FDD	Frequency division duplex
HBS	High band simultaneous
HLOS	High-level operating system
HMX	Qualcomm® Hexagon™ Matrix eXtensions
HTP	Qualcomm® Hexagon™ Tensor Processor
HVX	Qualcomm® Hexagon™ Vector eXtensions
ISP	Image Signal Processor
LPAI	Low-Power AI
NSA	Non-standalone
NTN	Non-Terrestrial Network
PoP	Package-on-package
QSH	Qualcomm® Sensing Hub
SA	Standalone
TDD	Time division duplex
TD-SCDMA	Time division synchronous code division multiple access
ULCA	Uplink carrier aggregation
UWB	Ultra wideband
VPU	Video processing unit
WLAN	Wireless local area network

Qualcomm Confidential – May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

Questions?

For additional information or to submit technical questions, go to: https://www.qualcomm.com/support

Qualcomm
Confidential - May Contain Trade Secrets
2026-02-27 05:38:47 GMT
hyungchul.won@lge.com

LEGAL INFORMATION

Your access to and use of this material, along with any documents, software, specifications, reference board files, drawings, diagnostics and other information contained herein (collectively this "Material"), is subject to your (including the corporation or other legal entity you represent, collectively "You" or "Your") acceptance of the terms and conditions ("Terms of Use") set forth below. If You do not agree to these Terms of Use, you may not use this Material and shall immediately destroy any copy thereof.

1) Legal Notice.

This Material is being made available to You solely for Your internal use with those products and service offerings of Qualcomm Technologies, Inc. ("**Qualcomm Technologies**"), its affiliates and/or licensors described in this Material, and shall not be used for any other purposes. If this Material is marked as "**Qualcomm Internal Use Only**", no license is granted to You herein, and You must immediately (a) destroy or return this Material to Qualcomm Technologies, and (b) report Your receipt of this Material to qualcomm.support@qti.qualcomm.com. This Material may not be altered, edited, or modified in any way without Qualcomm Technologies' prior written approval, nor may it be used for any machine learning or artificial intelligence development purpose which results, whether directly or indirectly, in the creation or development of an automated device, program, tool, algorithm, process, methodology, product and/or other output. Unauthorized use or disclosure of this Material or the information contained herein is strictly prohibited, and You agree to indemnify Qualcomm Technologies, its affiliates and licensors for any damages or losses suffered by Qualcomm Technologies, its affiliates and/or licensors for any such unauthorized uses or disclosures of this Material, in whole or part.

Qualcomm Technologies, its affiliates and/or licensors retain all rights and ownership in and to this Material. No license to any trademark, patent, copyright, mask work protection right or any other intellectual property right is either granted or implied by this Material or any information disclosed herein, including, but not limited to, any license to make, use, import or sell any product, service or technology offering embodying any of the information in this Material.

THIS MATERIAL IS BEING PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESSED, IMPLIED, STATUTORY OR OTHERWISE. TO THE MAXIMUM EXTENT PERMITTED BY LAW, QUALCOMM TECHNOLOGIES, ITS AFFILIATES AND/OR LICENSORS SPECIFICALLY DISCLAIM ALL WARRANTIES OF TITLE, MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, SATISFACTORY QUALITY, COMPLETENESS OR ACCURACY, AND ALL WARRANTIES ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. MOREOVER, NEITHER QUALCOMM TECHNOLOGIES, NOR ANY OF ITS AFFILIATES AND/OR LICENSORS, SHALL BE LIABLE TO YOU OR ANY THIRD PARTY FOR ANY EXPENSES, LOSSES, USE, OR ACTIONS HOWSOEVER INCURRED OR UNDERTAKEN BY YOU IN RELIANCE ON THIS MATERIAL.

Certain product kits, tools and other items referenced in this Material may require You to accept additional terms and conditions before accessing or using those items.

Technical data specified in this Material may be subject to U.S. and other applicable export control laws. Transmission contrary to U.S. and any other applicable law is strictly prohibited.

Nothing in this Material is an offer to sell any of the components or devices referenced herein.

This Material is subject to change without further notification.

In the event of a conflict between these Terms of Use and the *Website Terms of Use* on www.qualcomm.com , the *Qualcomm Privacy Policy* referenced on www.qualcomm.com, or other legal statements or notices found on prior pages of the Material, these Terms of Use will control. In the event of a conflict between these Terms of Use and any other agreement (written or click-through, including, without limitation any non-disclosure agreement) executed by You and Qualcomm Technologies or a Qualcomm Technologies affiliate and/or licensor with respect to Your access to and use of this Material, the other agreement will control.

These Terms of Use shall be governed by and construed and enforced in accordance with the laws of the State of California, excluding the U.N. Convention on International Sale of Goods, without regard to conflict of laws principles. Any dispute, claim or controversy arising out of or relating to these Terms of Use, or the breach or validity hereof, shall be adjudicated only by a court of competent jurisdiction in the county of San Diego, State of California, and You hereby consent to the personal jurisdiction of such courts for that purpose.

2) Trademark and Product Attribution Statements.

Qualcomm is a trademark or registered trademark of Qualcomm Incorporated. Arm is a registered trademark of Arm Limited (or its subsidiaries) in the U.S. and/or elsewhere. The Bluetooth® word mark is a registered trademark owned by Bluetooth SIG, Inc. Other product and brand names referenced in this Material may be trademarks or registered trademarks of their respective owners.

Snapdragon and Qualcomm branded products referenced in this Material are products of Qualcomm Technologies, Inc. and/or its subsidiaries. Qualcomm patented technologies are licensed by Qualcomm Incorporated.

Qualcomm Confidential - May Contain Trade Secrets
2026-03-27 05:38:47 GMT
kyungchul.kim@lge.com